Web Technologies – Assignment 1 – Project Proposal Template

GRIFFITH COLLEGE DUBLIN

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Signed: Mikhail, Vladimir, Georgii, Artem **Date: 09.03.2026**

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BEPRO

Introduction

This project proposes the development of a web-based appointment booking application called "BePro" that allows customers to schedule appointments with service providers such as barbershops, driving instructors, and tutors. The system will allow users to create accounts, browse available time slots, and book appointments. Service providers will be able to manage their schedules and track bookings through a dedicated dashboard.

The purpose of this document is to outline the proposed system, its key functionalities, user research, design decisions, and planned development timeline. It also includes initial user interface designs such as site maps and wireframes which will guide the development of the final application.

Functionality

Overview of Core Functionalities

The BePro application will deliver six core functionalities that address the needs of both customers and service providers. Each functionality has been designed to ensure a seamless user experience while meeting the technical requirements of the MERN stack, including client-side and server-side logic, multi-user support, and data persistence.

Detailed Functional Requirements

Functionality 1: User Registration and Authentication

Purpose:
To provide secure access to the application for both customers and service providers, ensuring that each user has a private, personalised account.

Description:
The system will support two user roles: customers and service providers. Users can register using their email address or via Google OAuth for faster onboarding. During registration, users will provide essential information such as name, email, and role. The system will securely store credentials using encryption (bcrypt) and manage session tokens via JWT (JSON Web Tokens).

How it works:

- **Client-side:** Users interact with a registration/login form. Input validation ensures email format is correct and passwords meet security criteria. Google login option simplifies the process and auto-fills profile information.
- **Server-side:** The server validates credentials, checks for existing users, hashes passwords, and stores user data in MongoDB. Upon successful authentication, a JWT is generated and sent to the client to maintain session state.

User benefits:
Secure, personalised access; saved preferences; ability to manage appointments and schedules across multiple sessions.

Supports multiple users and sessions:
Each user receives a unique JWT, ensuring sessions are isolated. Users can remain logged in across different devices or browsers without interfering with each other's sessions.

Functionality 2: Service Provider Search and Filtering

Purpose:
To help customers easily discover service providers (e.g., barbers, tutors, driving instructors) based on their needs and preferences.

Description:
Customers can browse a list of service providers, filter by category (e.g., barber, tutor), location, or rating, and view basic provider information such as bio, services offered, and availability.
How it works:
- **Client-side:** Users enter search criteria or select filters. The interface dynamically updates as filters are applied, providing instant feedback. Auto-suggestions may appear as users type location or service names.
- **Server-side:** The server processes search queries, filters provider records stored in MongoDB, and returns relevant results. Complex queries may involve geolocation-based sorting or rating-based ranking.

User benefits:
Quickly find the right provider without manually browsing through irrelevant listings.
Supports multiple users and sessions:
Search queries are independent per user session, and results are generated in real-time without cross-user interference.

Functionality 3: Appointment Booking
Purpose:
To enable customers to book available time slots with their chosen service provider efficiently and reliably.
Description:
After selecting a provider, customers can view a calendar of available time slots. They select a date and time, confirm the booking, and receive immediate confirmation. The system prevents double-booking by locking selected slots during the booking process.
How it works:
- **Client-side:** Users interact with a visual calendar interface. When a slot is selected, a booking form appears with optional notes (e.g., specific requests). Client-side validation ensures all required fields are completed.
- **Server-side:** Upon submission, the server checks slot availability in the database to prevent conflicts. If available, the booking is saved, and the provider's schedule is updated. A confirmation email is triggered via Node.js email service.

User benefits:
Hassle-free booking with real-time availability and instant confirmation.
Supports multiple users and sessions:
Concurrent booking attempts for the same slot are handled via atomic database operations, ensuring only one user can successfully book a given slot.

Functionality 4: Appointment Management for Customers
Purpose:
To allow customers to view, reschedule, or cancel their upcoming appointments from a central dashboard.
Description:
Customers have access to a personalised dashboard displaying all their upcoming and past appointments. They can click on any upcoming booking to view details, reschedule to a new available slot, or cancel if necessary.
How it works:
- **Client-side:** The dashboard retrieves and displays appointments from the server. Rescheduling triggers a new availability check, and cancellation prompts a confirmation dialog to prevent accidental deletions.

- **Server-side:** The server processes requests to update or delete appointment records. When an appointment is cancelled, the corresponding time slot is freed for other customers. Email notifications are sent to both customer and provider.

User benefits:
Full control over bookings; flexibility to adapt to changing plans.
Supports multiple users and sessions:
Each customer sees only their own appointments, and updates are reflected immediately across all active sessions.

Functionality 5: Schedule Management for Service Providers

Purpose:
To empower service providers to manage their availability, view upcoming bookings, and maintain control over their business schedule.
Description:
Service providers have a dedicated dashboard where they can set recurring availability (e.g., working hours), block out specific times, and view a calendar of confirmed appointments. They can also cancel appointments if necessary.
How it works:

- **Client-side:** Providers use an interactive calendar to define available time blocks. The interface prevents overlapping availability and visually distinguishes booked versus free slots.
- **Server-side:** Availability settings are stored in MongoDB and referenced during customer booking attempts. When a booking is made or cancelled, the provider's calendar updates in real-time.

User benefits:
Streamlined schedule management; reduced administrative workload; elimination of double-bookings.
Supports multiple users and sessions:
Each provider manages their own schedule independently. Changes to availability are instantly applied, affecting future booking possibilities for all customers.

Functionality 6: Dashboard Overview and Notifications

Purpose:
To provide both customers and service providers with a personalised summary of key information and alerts.
Description:
Upon logging in, users are greeted with a dashboard tailored to their role. Customers see upcoming appointments, recent bookings, and notifications (e.g., reminders, confirmations). Service providers see today's schedule, pending bookings, and alerts about cancellations or new bookings.
How it works:

- **Client-side:** The dashboard fetches and displays aggregated data from the server. Notifications appear as badges or pop-ups, with links to relevant sections.
- **Server-side:** The server aggregates data from multiple collections (appointments, users, notifications) and returns a structured response. Background processes may trigger notifications based on events (e.g., 24-hour reminder before an appointment).

User benefits:
At-a-glance awareness of upcoming commitments; timely alerts to stay informed.
Supports multiple users and sessions:
Dashboards are user-specific and update dynamically. Notifications are marked as read per session, ensuring a consistent experience across devices.

Summary of Multi-User and Multi-Session Support

The BePro application is designed from the ground up to support concurrent users and multiple sessions:

- **Authentication:** JWT-based sessions ensure each user's identity is securely maintained.
- **Data Isolation:** Database queries always filter by user ID, ensuring users only access their own data.
- **Concurrency Control:** Atomic database operations prevent race conditions during booking conflicts.
- **Real-time Updates:** Changes made in one session (e.g., booking an appointment) are immediately reflected when the same user accesses the system from another device.

This architecture guarantees that multiple users can interact with the system simultaneously without data corruption or session interference.

1 Research

User Analysis:

Understanding the needs, behaviors, and expectations of the primary user groups that will interact with the BePro system is made easier with the use of user analysis. Customers and service providers were determined to be the two main user groups for this project.

Customers:

Clients are those who wish to schedule appointments with various service providers, like tutors, driving instructors, and barbers. These consumers typically seek a quick and easy method for scheduling a time slot and verifying availability.

Characteristics:

- Age range: approximately 18 – 60.
- Regular users of smartphones and web applications
- Prefer quick and convenient booking systems
- Value clear schedules and instant confirmation

User needs:

- Easy account registration and login
- Ability to browse available services and providers
- View available time slots
- Book and manage appointments
- Receive booking confirmations

Service Providers

Professionals that provide services are known as service providers, and they require an effective means to schedule their appointments.

Examples:

- Barbers
- Driving instructors
- Tutors

Key characteristics:

- Self-employed or small business owners
- Need a simple scheduling system
- Want to avoid double bookings
- Need visibility of upcoming appointments

User needs:

- Create and manage their profile
- Set available working hours
- View upcoming bookings
- Manage or cancel appointments

Understanding these user groups helps guide the design and functionality of the BePro system.

Personas:

Personas are made-up characters that stand in for common system users. They assist designers in comprehending consumer requirements and creating solutions that deal with actual situations.

Persona 1: The Busy Student

Name: Sarah O'Connor

Age: 22

Occupation: University Student

Background:

Sarah works part-time and attends university, so her schedule is hectic. Rather than calling companies, she would rather make service reservations online.

Goals:

- Quickly find available appointments
- Book services without making phone calls
- Receive reminders about appointments

Frustrations:

- Businesses that require phone booking
- Unclear availability
- Long booking processes

How BePro Helps:

BePro allows Sarah to quickly view available time slots and book appointments online in just a few clicks.

Persona 2: The Independent Service Provider

Name: Michael Byrne

Age: 35

Occupation: Driving Instructor

Background:

Michael works independently and schedules his driving lessons manually using phone calls and a notebook.

Goals:

- Organise his schedule more efficiently
- Avoid double bookings
- Easily see upcoming appointments

Frustrations:

- Managing bookings manually
- Last-minute cancellations
- Forgetting scheduled appointments

How BePro Helps:

BePro provides Michael with a dashboard where he can manage his schedule and track bookings easily.

Competitive Analysis:

A competitive study looks at comparable systems to determine their advantages and disadvantages. This makes it easier to find areas where the BePro platform can be improved.

1. Booksy

Booksy is a popular appointment booking platform used mainly by barbershops and beauty professionals.

Strengths:

- Easy-to-use interface
- Mobile application
- Automatic reminders

Weaknesses:

- Mainly focused on beauty services
- Service providers must pay subscription fees

2. Calendly

Calendly is a widely used scheduling tool for meetings and appointments.

Strengths:

- Simple scheduling system
- Integration with calendars
- Automatic scheduling

Weaknesses:

- Designed mainly for business meetings
- Less suitable for service-based businesses like barbers or tutors

3. Fresha

Fresha is another booking system used in beauty and wellness industries.

Strengths:

- Online booking
- Staff scheduling
- Payment integration

Weaknesses:

- Focused mostly on salons and spas
- Can be complex for small independent providers

Opportunity for BePro

BePro wants to offer an easy-to-use booking platform that can be used for a variety of services, including driving instructors, tutors, and barbers.

Content Inventory:

A content inventory lists all the pages and content elements that will be included in the application.

1.1.1 Public Pages

- Home Page
- About Page
- Login Page
- Registration Page
- Service Provider Listings
- Appointment Booking Page

1.1.2 Customer Dashboard

- User Profile

- My Appointments
- Booking History
- Notifications

1.1.3 Service Provider Dashboard

- Provider Profile
- Schedule Management
- Appointment Requests
- Booking History
- Availability Settings

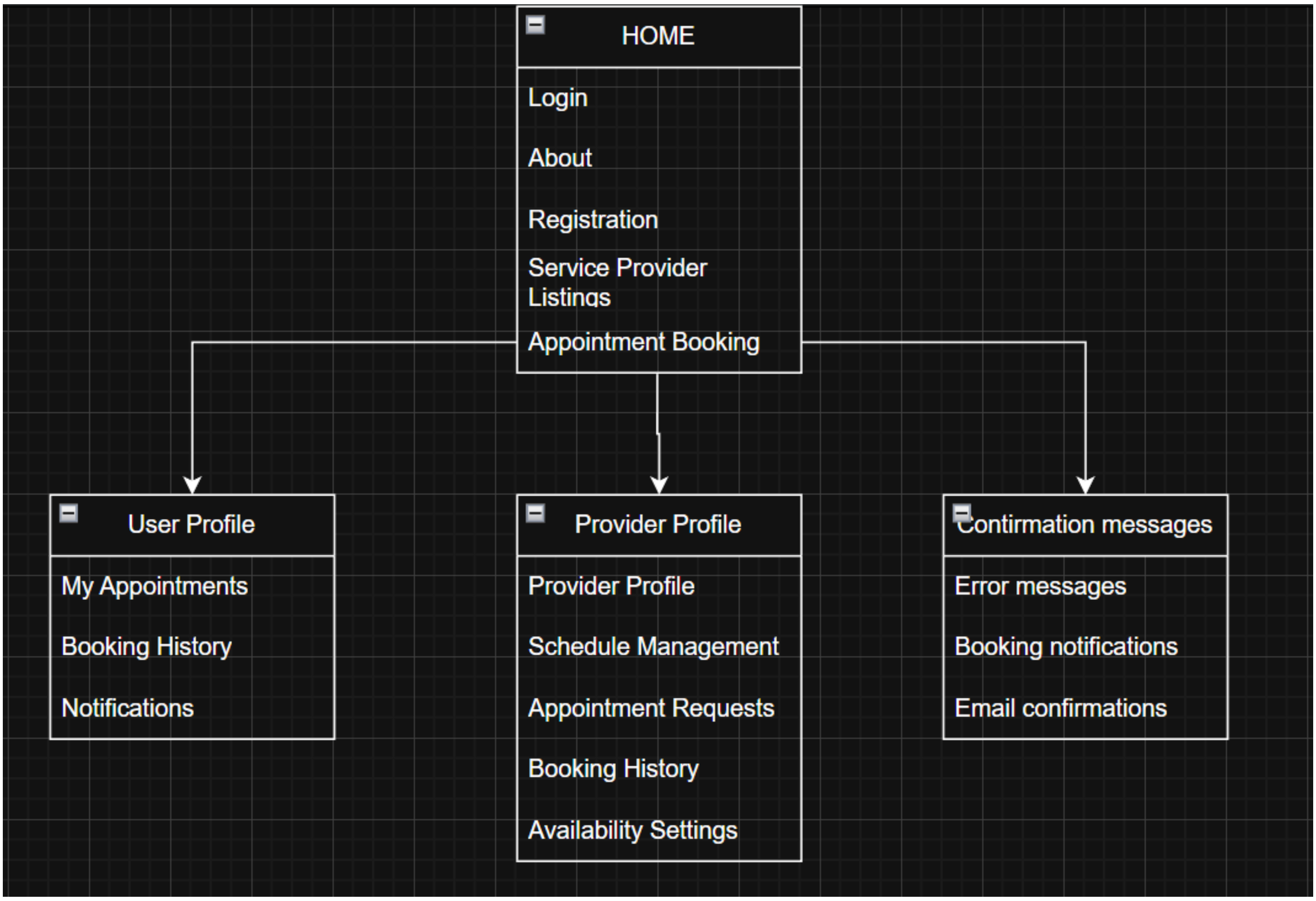
1.1.4 Other System Content

- Confirmation messages
- Error messages
- Booking notifications
- Email confirmations

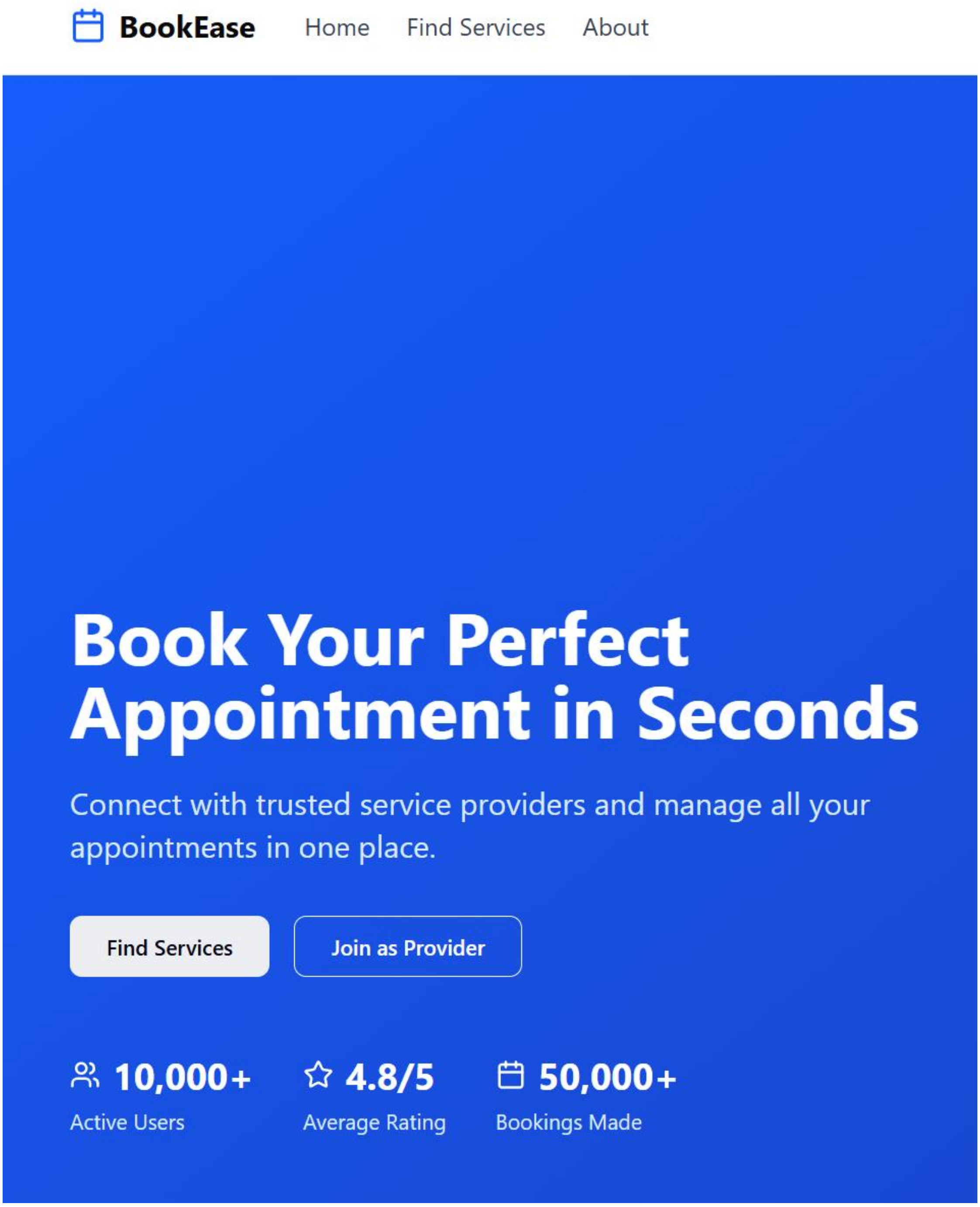
This content inventory helps ensure that all required pages and features are identified before development begins.

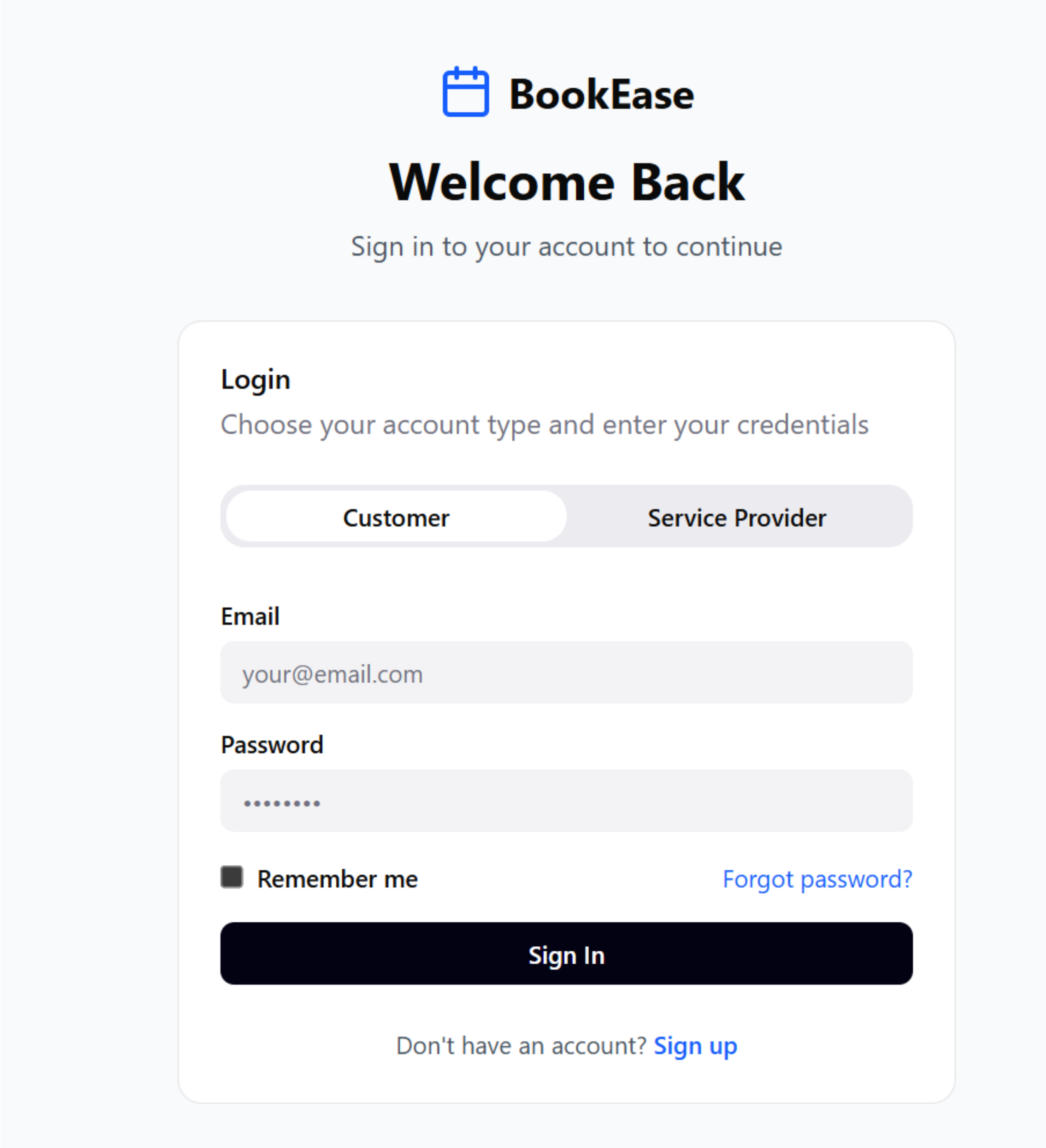
2 Design specifications

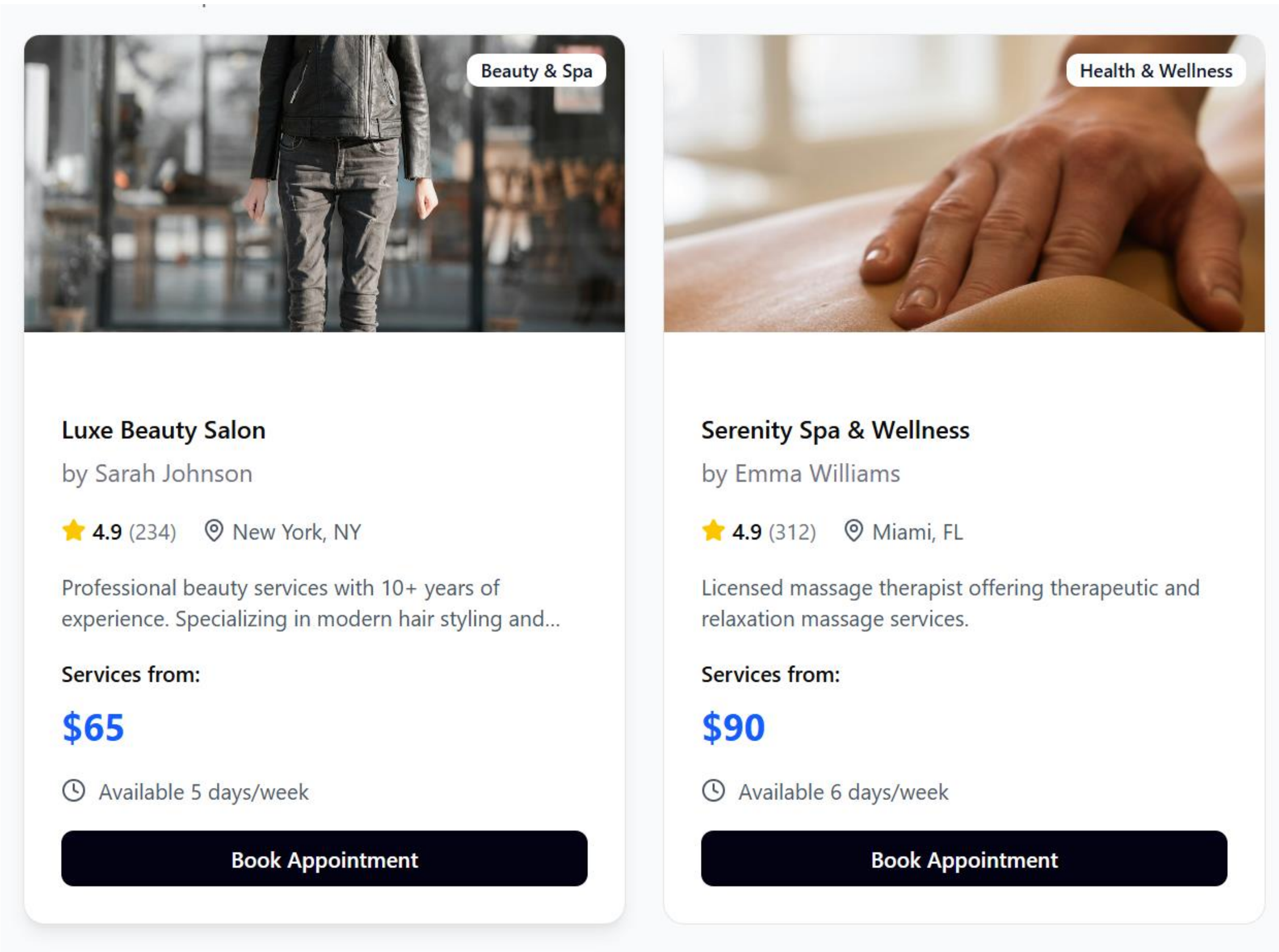
2.1 Site map



2.2 Wireframe templates




BookEase
Welcome Back
Sign in to your account to continue
Login
Choose your account type and enter your credentials
Customer
Service Provider
Email
your@email.com
Password
••••••••
Remember me
Forgot password?
Sign In
Don't have an account? Sign up



2.3 Colour

White #ffffff

Blue #1E4BDD

Grey #D3D3D3

2.4 Fonts

Playfair Display

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5 The project timeline/ schedule

We intend to work on the development and advancement of our project for approximately one hour every day of the week. To enable everyone to work together and contribute to the project, we will first build a repository on GitHub and add all participating developers.

Frontend development will be the initial phase. We anticipate working on the application's user interface and primary graphic elements for two to three weeks.

We will next proceed to the backend development. This section is a little more complicated because it necessitates careful consideration of several factors, including the database structure, the DTO layer, and security features like user registration and login. Our goal is to finish this phase in about a month.

We anticipate that deployment, the last phase, will take roughly a week.

Each team member will devote an equal amount of effort to the project, and all jobs will be divided equally among them.